SMART MANUFACTURING AI

Visual Quality Inspection & Automated Defect Intelligence Platform

RESULT: FAIL

FAILED QUALITY AUDIT

Inspection ID:	INSP-1793369862-7C3A41	**Timestamp:**	2026-10-30 19:47:46
Component Name:	Gearbox casing	**Similarity Score (SSIM):**	99.38%
Total Defect Regions:	2	**Affected Surface Area:**	1.046%
High-Severity Defects:	0	**Processing Duration:**	3712.1 ms

Visual Evidence & Comparative Computer Vision Analysis

Figure A: Reference (Good Component)

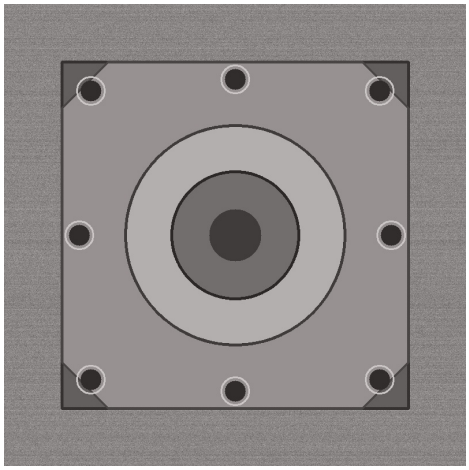


Figure B: Inspected Component

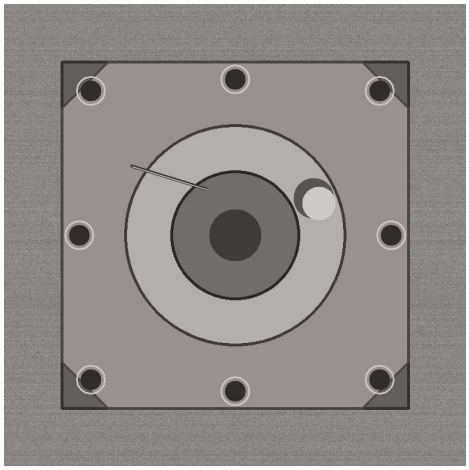


Figure C: SSIM Difference Heatmap

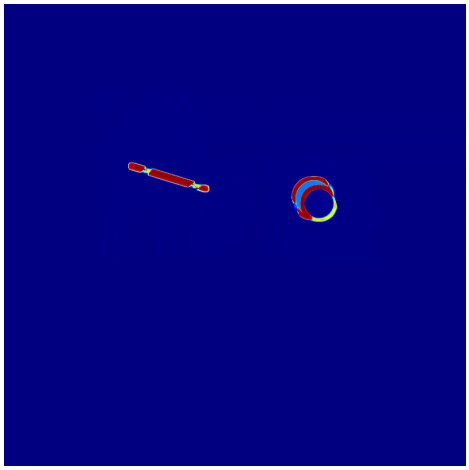
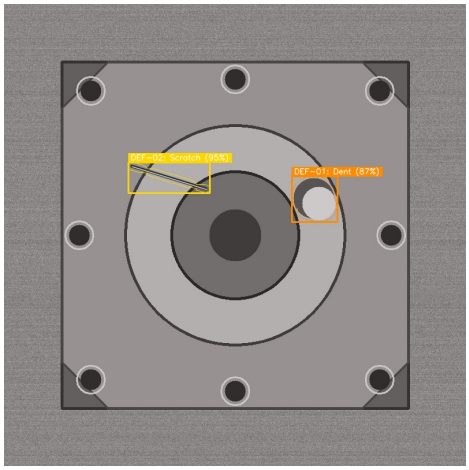


Figure D: Defect Highlighted & Annotated



Detected Defects & Anomaly Registry

Defect ID	Category	Severity	Score	Location (X, Y, W, H)	Area (% / px)	Description

DEF-01	**Dent**	**MEDIUM**	0.87	(498, 298) 79x79	0.75% (4823px)	Localized plastic deformation / depression (0.75% surface area).
DEF-02	**Scratch**	**LOW**	0.95	(215, 274) 141x53	0.29% (1872px)	Superficial hairline scratch on component surface.

AUTOMATED QUALITY ENGINEERING RECOMMENDATIONS:

- Verify whether surface dent impacts critical mating face sealing or bolt seat flatness.
- Check material handling conveyors and robotic gripper pad pressure to prevent impact indentation.
- Inspect if scratch depth exceeds allowable paint/protective primer specification (e.g., > 25 microns).
- Route to rework station for deburring, light polishing, and primer re-touch.

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